

1. Block identification

IP Block Name:	Window Comparator
IP Block Abbreviation:	WINCOMP
Category:	Analog IP — open-source library block
Target PDK:	IHP SG13CMOS5L (130 nm CMOS, 5 metal layers)
Design flow:	Xschem / ngspice / Magic / KLayout / Netgen (IIC-OSIC-TOOLS)
Submission:	Chipalooza Challenge #2 — Open Circuit Design
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2. Definition

A window comparator is a dual-threshold voltage detector. It compares a single input voltage (VIN) against two independent reference voltages, VL (lower threshold) and VH (upper threshold), and asserts a digital output flag when the input lies strictly between the two thresholds.

3. Utilization / target applications

- Battery over-voltage / under-voltage monitoring
- Power-good / brown-out detection in SoC power management
- Sensor out-of-range flagging (analog level supervision)
- Simple threshold-based wake-up or alarm logic in mixed-signal chips

4. Pin list (I/O)

Pin	Direction	Type	Description
VIN	Input	Analog	Signal to be monitored
VL	Input	Analog	Lower threshold reference (external)
VH	Input	Analog	Upper threshold reference (external)
VDD	Power	Supply	1.2 V core supply
VSS	Power	Ground	0 V
OUT	Output	Digital	High when $VL < VIN < VH$; low otherwise

5. Electrical specifications

Target design values for a single-stage 5T differential comparator in 130 nm CMOS (SG13CMOS5L).

Parameter	Symbol	Min	Typ	Max	Unit
Supply voltage	VDD	1.08	1.2	1.32	V
Operating temperature	TEMP	−40	27	125	°C
Input common-mode range	ICMR	0.3	—	VDD−0.3	V
Differential DC gain (input stage, pre-buffer)	AVDC	30	40	50	dB
Input-referred offset voltage (3σ)	VOS	—	±5	±20	mV
Common-mode rejection ratio	CMRR	45	55	—	dB
Power-supply rejection ratio	PSRR	35	45	—	dB
Hysteresis (if implemented)	VHYS	—	15	—	mV
Propagation delay, 10 mV overdrive	tPD	—	10	20	ns
Static power, both comparators	PSTAT	—	15	30	μW
Output logic high	VOH	VDD−0.1	VDD	—	V
Output logic low	VOL	—	0	0.1	V
Output rise/ fall time (10-90%)	tR	—	1.5	3	ns
Output load	CL	—	—	TBD	pF
Active area, both comparators + AND gate	AREA	—	TBD	—	mm ²

7. Power consumption

Parameter	Symbol	Min	Typ	Max	Unit	Conditions
Static bias current, per comparator	IQ	—	5	—	μA	quiescent, VIN away from either threshold
Static power, both comparators + digital gates	PSTAT	—	15	30	μW	VDD = 1.2 V, TEMP = 27 °C
Dynamic switching energy per output edge	ESW	—	TBD	—	pJ	$\approx CL \times VDD^2$, fill in once CL is set

For a slowly-varying monitored signal (the typical use case — battery or sensor level monitoring), OUT toggles rarely, so static bias current dominates total power draw; dynamic switching energy only matters if OUT is expected to transition frequently.

8. Architecture notes

- Two matched single-stage 5T differential comparators (one vs. VL, one vs. VH) followed by a standard-cell AND gate combining their outputs into OUT.
- VL and VH are external reference pins — no on-chip bandgap or resistor ladder in this block; reference generation is left to the system integrator.
- Both comparator instances share one matched analog subcircuit (common- centroid layout) to minimize offset and layout effort.
- The proposed comparator architecture is designed as a high-precision, open-loop 5T differential pair, intentionally implemented without hysteresis to ensure maximum input sensitivity and measurement accuracy. However, the design is modular; provisions for programmable hysteresis have been architecturally reserved, allowing for its integration during the layout phase should system-level testing indicate a requirement for improved noise immunity.

9. Test plan

9.1 DC characterization

- **Threshold / offset (VOS):** DC sweep VIN from 0 to VDD with VL, VH at nominal values; record the VIN value where OUT transitions for each comparator and compare it to the ideal VL/VH.
- **DC gain (AVDC):** OUT is digital (saturated), so extract gain from the internal pre-buffer analog node — bias the comparator at its trip point (.op), then run a small-signal .ac analysis around that point.
- **CMRR:** sweep VIN and the active reference (VL or VH) together at a fixed differential offset across the full common-mode range; measure the resulting shift in trip point.
- **PSRR:** sweep VDD across its min-max range at fixed VIN/VL/VH; measure the resulting shift in trip point.
- **ICMR:** confirm correct switching with VIN swept across its full common-mode range, at both supply corners.
- **Corners:** repeat the threshold/offset sweep across process corners (SS, FF, SF, FS, TT), temperature (-40, 27, 125 °C), and supply (min, typ, max).
- **Monte Carlo:** mismatch + process Monte Carlo (200-500 points) on the threshold sweep to get the statistical VOS distribution (3σ).

9.2 AC / transient characterization

- **Propagation delay:** step the input across each threshold at a defined overdrive (e.g. ± 10 mV, ± 50 mV) with the target output load CL; measure tPLH/tPHL from the 50% input crossing to the 50% output crossing.
- **Rise/fall time:** measure 10-90% transition time on OUT for the same step stimulus.
- **Hysteresis:** apply a slow triangular/ramp input crossing a threshold in both directions; the difference between rising and falling trip points is VHYS.

9.3 Power characterization

- **Static power:** measure total supply current with VIN held away from either threshold (no switching), across process/temperature/supply corners.
- **Dynamic energy:** measure the supply current transient during a single output edge and integrate to get energy per transition.

9.4 Layout-dependent verification

- DRC clean against IHP SG13CMOS5L layout rules.
- LVS clean (schematic vs. layout).
- Re-run sections 9.1-9.3 on the PEX-extracted post-layout netlist and confirm no significant degradation vs. schematic-level results.

9.5 Toolchain

Xschem for circuit , ngspice for DC/AC/transient/corner/Monte Carlo simulation, Magic + KLayout for layout, Netgen for LVS.

10. Designer CV

The CV is attached with the proposal email